

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458957.6587 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.027
Transit depth (Rp/Rs)^2	2.43 +/- 0.84 [%]
Orbital Inclination (inc)	82.21 +/- 2.11
Airmass coefficient 1 (a1)	0.88 +/- 0.011
Airmass coefficient 2 (a2)	0.093 +/- 0.076
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#1 - [605, 353]
Optimal Aperture	3.36
Optimal Annulus	13.83
Transit Duration (day)	0.0489 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.027 vs 0.1594 (deviation 0.13 σ)
Duration fit vs published	0.0489 d vs 0.0512 d (deviation 0.34 σ)
Mid-time O-C	-4.6 min
Depth SNR	5.8
Residual scatter	1.29 %

Light curve

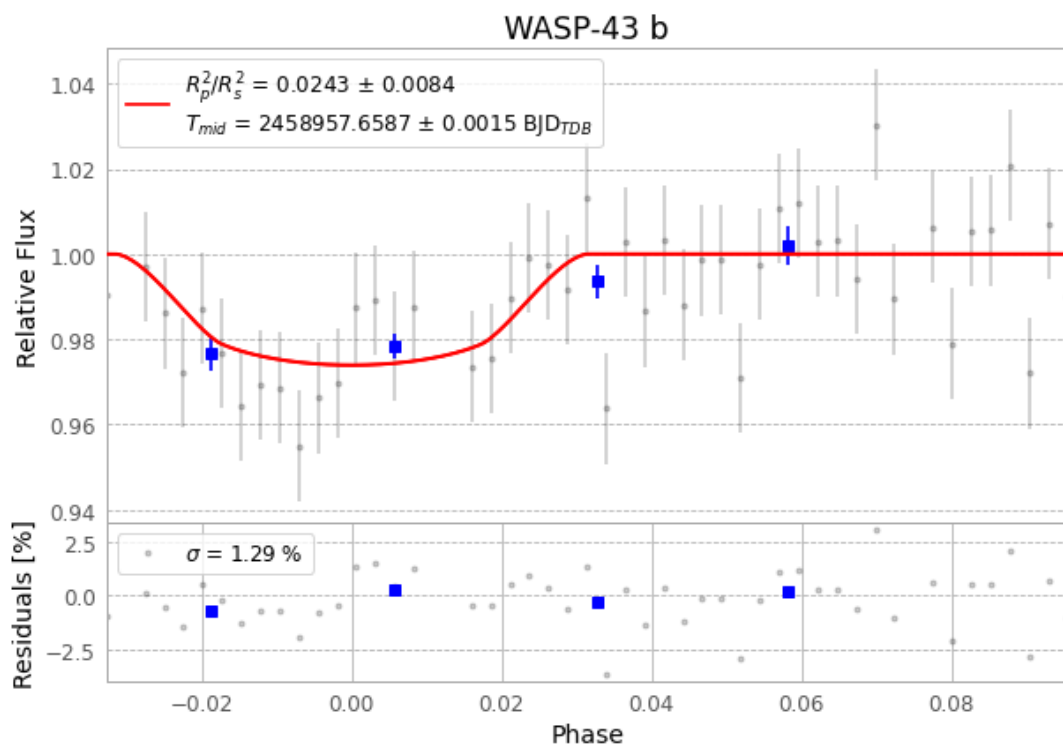


Figure 1 — FinalLightCurve_WASP-43 b_2020-04-17.png (SHA-256 381534a60e366fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f006fbb24957d873...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

50 FITS frames (33 MB), WASP-43200418030316.FITS → WASP-43200418053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200418.log (f8ee643c7383...), FinalLightCurve_WASP-43 b_2020-04-17.png (381534a60e36...), FinalLightCurve_WASP-43 b_2020-04-17.csv (083fedc672b7...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 15:37Z.